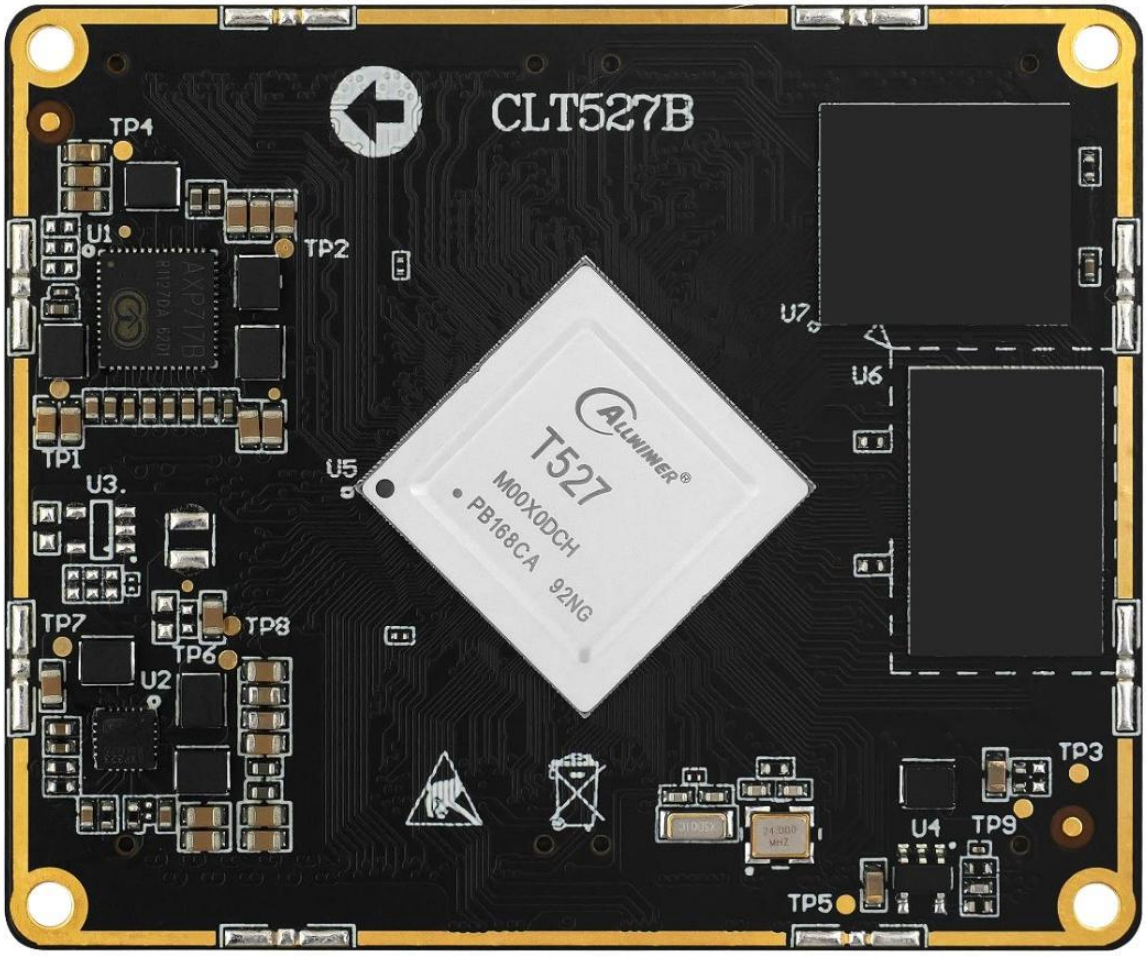
ATK-CLT527B

Core Board Specification

V1.0



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Revision History:

Version	Version Update Notes	Responsible person	Proofreading	Date
V1.0	release officially	ALIENTEK Linux Team	ALIENTEK Linux Team	2025.03.01

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Chapter 1. Core board overview

1.1 Core board introduction

The Core Board ATK-CLT527B by ALIENTEK is a high-performance embedded core board based on the domestic Allwinner T527 octa-core industrial-grade processor, providing rich hardware interfaces and complete software support.

The ATK-CLT527B core board adopts a BTB (board-to-board) connector for connection with the baseboard.

The core board is equipped with LPDDR4X high-speed memory and eMMC large-capacity storage as standard, meeting the storage capacity requirements of most products. Customized memory and storage capacities are also available according to batch demands.

The core board adopts the Allwinner T527 octa-core industrial-grade high-performance processor. The T527 is a new-generation high-performance and low-cost AI platform solution launched by Allwinner for the smart device market, mainly targeting the mid-to-high-end smart device sector.

The T527 series chips integrate multiple high-performance computing units, including an 8-core 64-bit Cortex-A55 CPU, G57 MC1 GPU, HiFi4 DSP, RISC-V MCU, and a 2 TOPS NPU (the NPU is **only supported by the T527M02X0DCH model**, and the core boards currently sold by our company do not use this variant). The CPU frequency can reach up to 1.8 GHz.

It features powerful hardware encoding and decoding capabilities, as well as rich display and camera interfaces. It supports common interfaces such as CAN, PCIe, SDIO, USB, GMAC, SPI, TWI (I2C), I2S, UART, ADC, and PWM.

It can be widely used in content sharing and self-service interactive terminals, industrial intelligence, industrial control, automotive electronics, intelligent manufacturing, and other electronic commercial and industrial equipment fields. Meanwhile, it supports operating systems including Linux, Android, and Debian to meet the requirements of complex applications.

The core board provides abundant software resources, including but not limited to Tina5.0 Linux SDK, Android 13 SDK, Linux Buildroot & Android 13 system firmware, Qt source code, relevant development tools and development environments, etc. Corresponding development documents are also provided, including development environment setup, SDK compilation, packaging & burning guide, hardware testing guide, baseboard hardware design reference, etc.

In addition, ALIENTEK has specially compiled a series of materials for core board users that are needed during the development phase, covering schematics, baseboard design reference documents, mechanical structures, component packages, connector specifications, Linux Buildroot & Android system firmware and SDK source code, compilers, software packages, etc., facilitating rapid development, improving development efficiency and shortening the development cycle.

ALIENTEK core board is supported by detailed and abundant documentation with strong practical operability, helping customers shorten the development cycle and accelerate product time-to-market.

1.2 Typical Application Areas

It can be widely used in content sharing and self-service interactive terminals, industrial intelligence, industrial control, automotive electronics, intelligent manufacturing, as well as other electronic commercial and industrial equipment fields.

Chapter 2. Core Board Resources

2.1 Hardware Resources

The ATK-CLT527B core board is based on the Allwinner T527 octa-core high-performance heterogeneous multi-core processor, with onboard LPDDR4X memory and EMMC storage, providing efficient data processing capability and large-capacity high-speed storage support for embedded systems. Meanwhile, the core board is equipped with a high-performance dedicated PMIC power management chip to meet the stable power supply requirements of high-performance systems.

Figure 2.1-1 Front Side of the Core Board

Figure 2.1-2 Back Side of the Core Board

2.2 Processor Resources

The block diagram of T527 processor parameters is shown in the figure below.

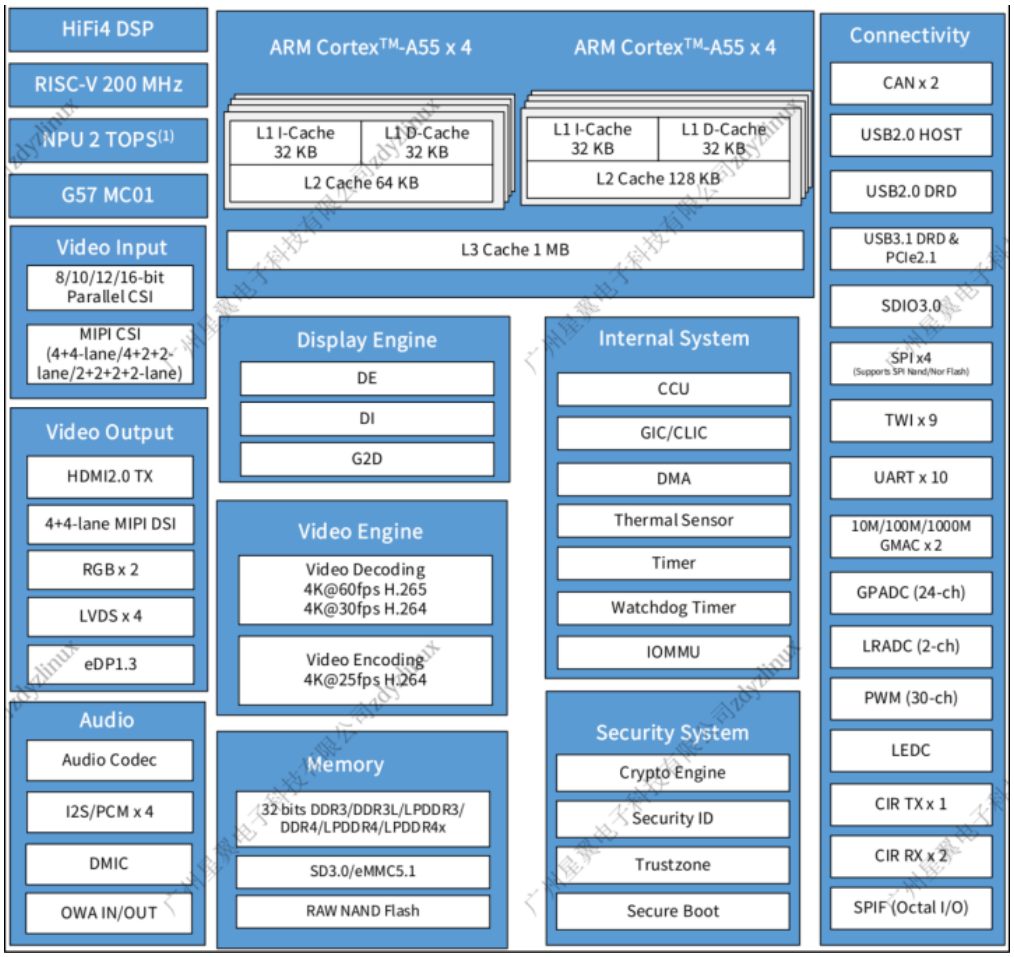


Figure 2.2-1 T527 Processor Block Diagram

The main internal resources of the T527 processor are listed in the table below. For more detailed parameters, please refer to the T527 processor datasheet.

T527 Chip Specifications	
CPU	8-core 64-bit ARM Cortex-A55, with a main frequency up to 1.8 GHz
GPU	ARM Mali-G57: · Supports OpenGL ES 3.2/2.0/1.1, Vulkan 1.1/1.2/1.3 and OpenCL 2.2
NPU	The T527 series chip is available in two models: T527M02X0DCH and T527M00X0DCH. Only the T527M02X0DCH model supports the NPU (2 TOPS). The core boards currently sold by our company are **not** equipped with this model. For details, please consult our technical support team!
MCU	RISC-V XuanTie E906, main frequency 200MHz
DSP	Cadence Tensilica HiFi4 DSP
G2D	2D Image Acceleration Engine: · Maximum resolution: 2560×2560 · Supports image rotation, mirroring, cropping and stitching, image data downsampling and stepless scaling, data format conversion, color space conversion, etc.
ISP	· Supports online mode and offline mode.

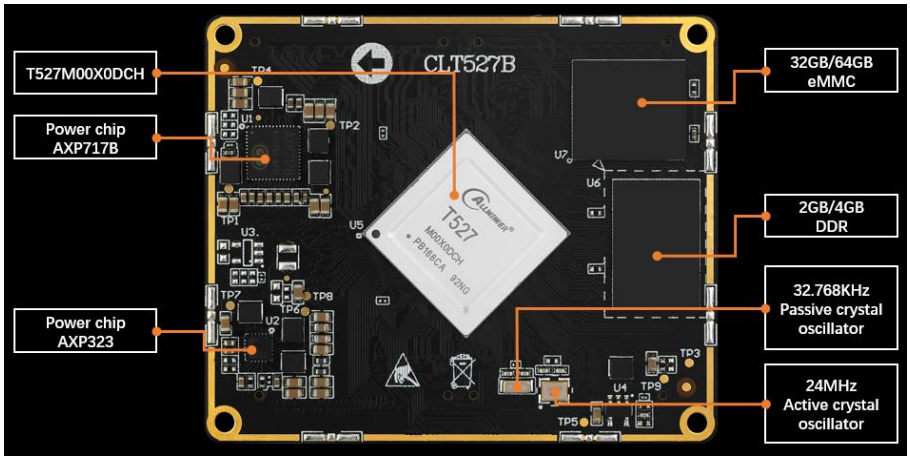
	· Maximum processing capability: 8M@30fps · Maximum input resolution in online mode: 3264×4224 · Data bit width: 8/10/12/14/16 bits
Video Encoding	· H.264 - 4K@25fps · MJPEG - 4K@15fps · JPEG - 8192*8192
Video Decoding	· H.265 - 4K@60fps · VP9 - 4K@60fps · H.264 - 4K@30fps · H.263/VP8/AVS/AVS+/MPEG-1/MPEG-2/MPEG-4 - 1080P@60fps · JPEG - 16384 * 16384
Video Input	Parallel CSI： · Supports 1 16-bit parallel CSI interface · Supports 8/10/12/16-bit width · Complies with BT.656, BT.601, BT.1120 interface standards · Maximum pixel clock: 148.5MHz
	MIPI CSI： · Supports up to 2 lanes × 4-lane / 4 lanes × 2-lane / 1 lane × 4-lane + 2 lanes × 2-lane · Supports up to 8M@30fps input, with resolution up to 3264(H)×2448(V)
Display (supports dual-screen different display)	HDMI2.0 OUT： · Supports up to 4K@60fps · Supports I2S
	MIPI DSI x2： · Supports 2 × 4-lane MIPI DSI interfaces · Supports Dual-Link · Supports up to 2560×1600@60fps (Dual-Link) and 1920×1200@60fps (Single-Link)
	RGB x2： · Supports up to 1920x1080@60fps
	LVDS x4： · Supports 2 Dual-Link LVDS interfaces · Supports up to 1366x768@60fps (Single-Link) and 1920x1080@60fps (Dual-Link)
	eDP1.3： · Supports 1 4-lane eDP TX interface · Supports up to 2.5K@60fps and 4K@30fps · Supports I2S
	Built-in Audio codec： · Supports 3 channels of audio output: LINEOUTLP/N, LINEOUTRP/N, HPOUTL/R · Supports 3 channels of audio input: MICIN1P/N, MICIN2P/N, MICIN3P/N
Audio	I2S/PCM x4：

	· Supports up to 16 channels, sampling rate from 8KHz to 384KHz, 8bit to 32bit
	DMIC： · Supports sampling rates from 8KHz to 48KHz
	OWA： · 1 OWA TX channel, 1 OWA RX channel · Compatible with S/PDIF standard interface
Storage	SD3.0
	RAW NAND Flash
	eMMC5.1
Other peripherals	USB2.0 OTG x1, supports firmware burning
	USB3.1 OTG x1, IO multiplexed with PCIe2.1 interface
	USB2.0 HOST x1
	GMAC x2, supports 10/100/1000 Mbps data rate and RGMII/RMII interfaces.
	PCIe2.1 x1： · Supports 1 PCIe2.1 x1 lane interface, 5Gbps/lane · IO multiplexed with USB3.1 interface · Supports RC mode only
	CAN x2, standard CAN interface (does not support CAN-FD)
	UART x10： · Maximum rate: 10 Mbit/s · Supports 5~8 data bits and 1/1.5/2 stop bits
	TWI (I2C) x9： · Compatible with standard I2C bus protocol · Supports 100 Kbit/s (Standard mode) and 400 Kbit/s (Fast mode) speeds
	SPI x4： · Maximum clock frequency: 100 MHz · Master/Slave mode configurable
	IR: Supports 2 IR RX channels and 1 IR TX channel
	PWM (30-ch): Supports pulse/cycle/complementary output and capture input
	ADC: Supports 24-ch GPADC, 2-ch LRADC
	GPIO x189
	SDIO3.0, SPIF（ Octal I/O）

Note: These are the resource parameter values from the **T527 processor datasheet**, not the available resource parameters for the core board.

The core board leads out most of the I/O pins on the processor. Users can design their own baseboard according to their needs to utilize the I/O resources on the core board and configure the I/O multiplexing to implement the required functions.

2.3 Hardware Parameters



The hardware parameters of the core board are shown in the table below.

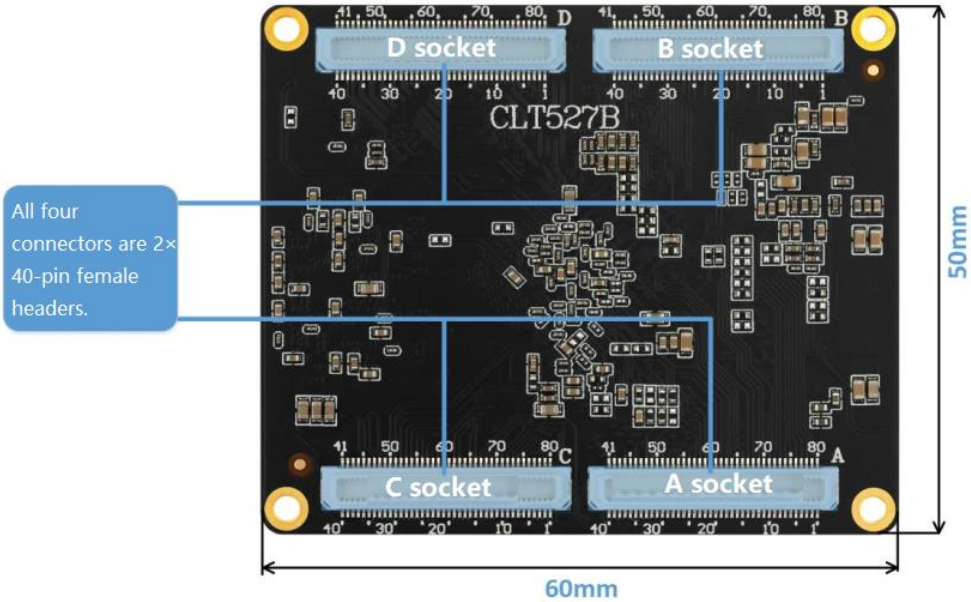
Item	Parameter	Remark
Core Board Model	ATK-CLT527B	
Dimensions	60mm*50mm	Length * Width
Processor Model	T527M00X0DCH	This model does not include an NPU.
Processor Architecture	8-core Cortex-A55 CPU, up to 1.8 GHz main frequency	Heterogeneous multi-core architecture, integrating RISC-V MCU, DSP, and GPU processing units.
Power Management	AXP717B + AXP323	Allwinner officially recommended power solution
Memory	2GB/4GB LPDDR4X	The following two configurations are available for selection: · 2GB DDR + 32GB eMMC · 4GB DDR + 64GB eMMC The model is subject to the actual components mounted. Bulk customization and replacement are also available.
Storage	32GB/64GB eMMC	
Operating Voltage	4V	To ensure stable operation, it is recommended to design the current at around 3A; the actual value shall be subject to measurement in the working scenario.

Power Consumption	⩾3.0W	The actual power consumption depends on the working scenario.
Operating Temperature	Commercial grade:0℃ ~ +70℃ Industrial grade：-40℃ ~ +85℃	The temperature range of the commercial-grade core board is actually wider than the rated value, and testing can be performed according to the actual requirements of the project.
Number of Interface Pins	320PIN（4*80）	
Pin Pitch	0.5mm	
Interface type	Four 80-pin anti-misplug BTB female connectors for board-to-board connection.	
Core board PCB to baseboard height	3mm	Height between mated core board connector and baseboard connector
Reserved mounting holes on core board	M2 fixing screws can be used.	There is one mounting hole at each of the four corners of the core board.
PCB technology	10-layer, gold-finger plating process, independent ground and signal layers.	Adopts lead-free process.

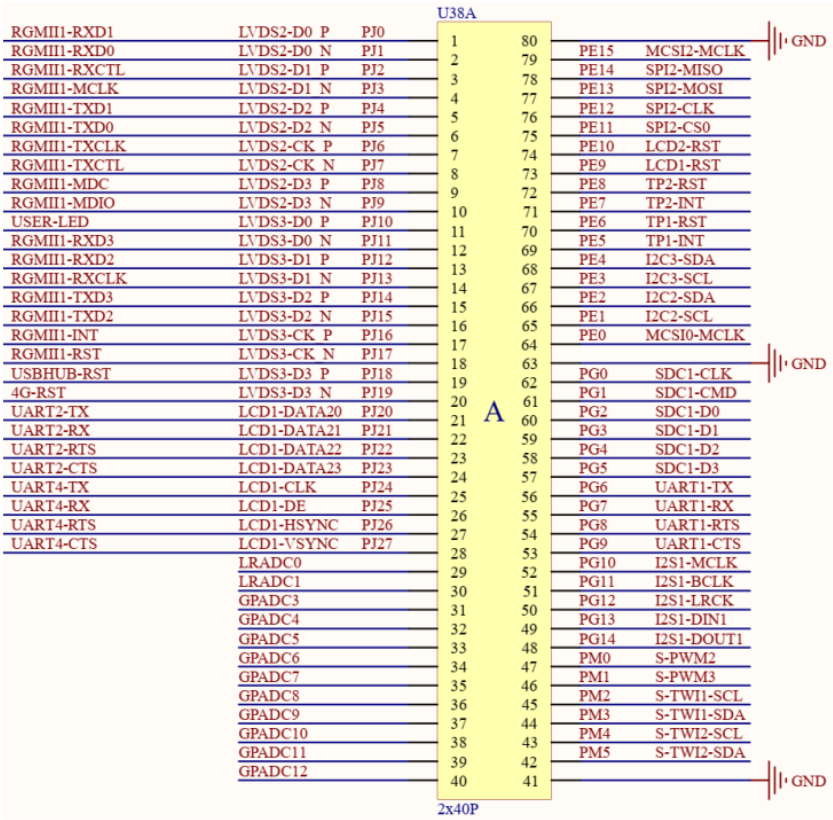
2.4 Pin Sequence and Interface Definition

The ATK-CLT527B core board is equipped with four 2×40-pin BTB connectors, leading out a total of 320 pins, including functional pins of the main control T527, GND, power supply pins, etc.

The four BTB connectors are named A, B, C, D respectively. There are silk-screen markings on the back of the core board, as well as pin number labels for easy reference by users, as shown in the figure below.

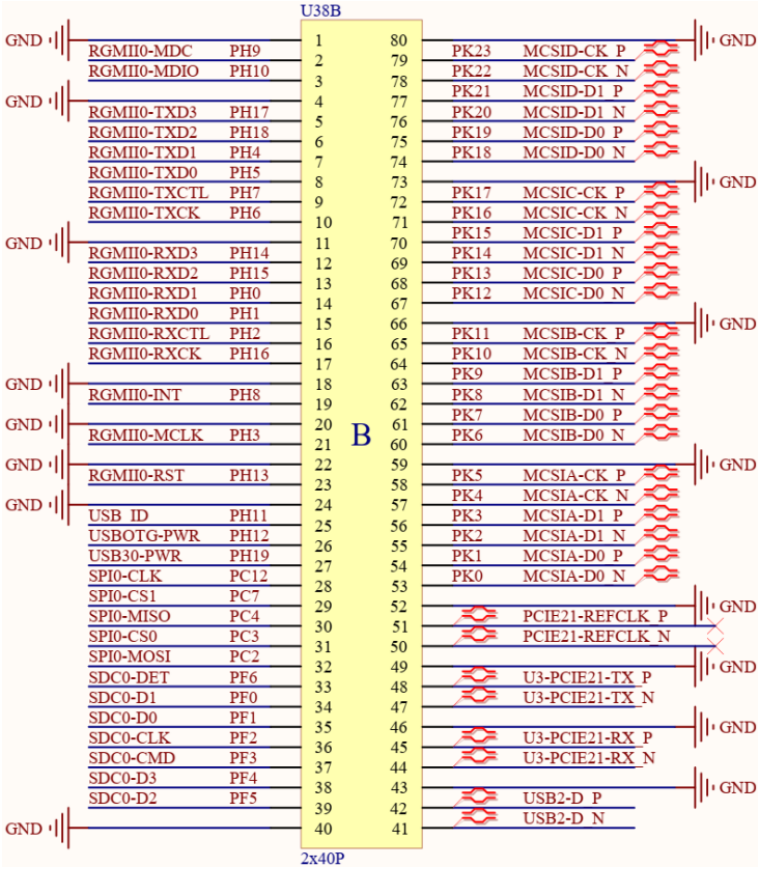


The specific pin function definitions are shown in the figure below:

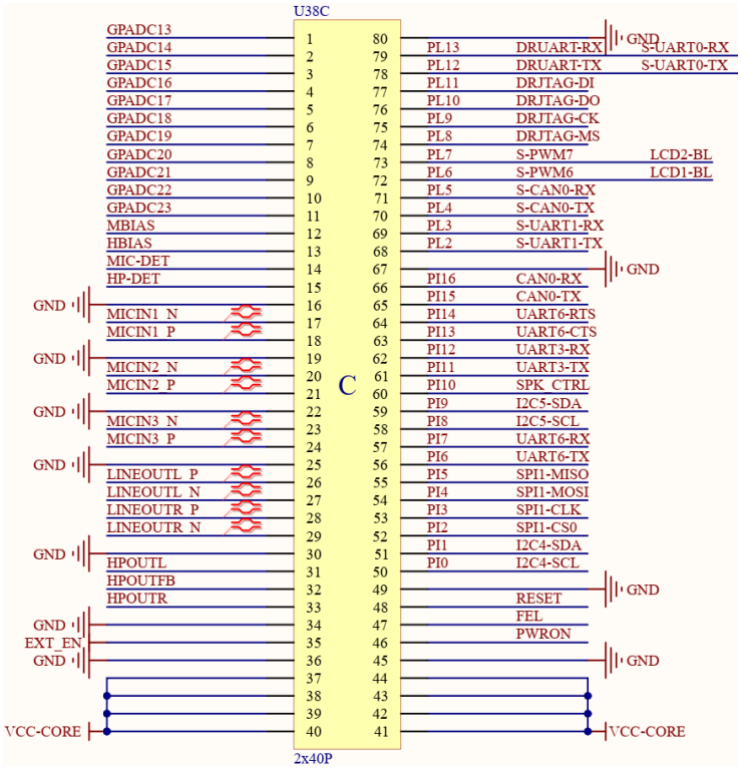


RGMII1-RXD1	LVDS2-D0_P	PJ0	1	80	PE15	MCSI2-MCLK	GND
RGMII1-RXD0	LVDS2-D0_N	PJ1	2	79	PE14	SPI2-MISO	
RGMII1-RXCTL	LVDS2-D1_P	PJ2	3	78	PE13	SPI2-MOSI	
RGMII1-MCLK	LVDS2-D1_N	PJ3	4	77	PE12	SPI2-CLK	
RGMII1-TXD1	LVDS2-D2_P	PJ4	5	76	PE11	SPI2-CS0	
RGMII1-TXD0	LVDS2-D2_N	PJ5	6	75	PE10	LCD2-RST	
RGMII1-TXCLK	LVDS2-CK_P	PJ6	7	74	PE9	LCD1-RST	
RGMII1-TXCTL	LVDS2-CK_N	PJ7	8	73	PE8	TP2-RST	
RGMII1-MDC	LVDS2-D3_P	PJ8	9	72	PE7	TP2-INT	
RGMII1-MDIO	LVDS2-D3_N	PJ9	10	71	PE6	TP1-RST	
USER-LED	LVDS3-D0_P	PJ10	11	70	PE5	TP1-INT	
RGMII1-RXD3	LVDS3-D0_N	PJ11	12	69	PE4	I2C3-SDA	
RGMII1-RXD2	LVDS3-D1_P	PJ12	13	68	PE3	I2C3-SCL	
RGMII1-RXCLK	LVDS3-D1_N	PJ13	14	67	PE2	I2C2-SDA	
RGMII1-TXD3	LVDS3-D2_P	PJ14	15	66	PE1	I2C2-SCL	
RGMII1-TXD2	LVDS3-D2_N	PJ15	16	65	PE0	MCSI0-MCLK	
RGMII1-INT	LVDS3-CK_P	PJ16	17	64			
RGMII1-RST	LVDS3-CK_N	PJ17	18	63	PG0	SDC1-CLK	GND
USBHUB-RST	LVDS3-D3_P	PJ18	19	62	PG1	SDC1-CMD	
4G-RST	LVDS3-D3_N	PJ19	20	61	PG2	SDC1-D0	
UART2-TX	LCD1-DATA20	PJ20	21	60	PG3	SDC1-D1	
UART2-RX	LCD1-DATA21	PJ21	22	59	PG4	SDC1-D2	
UART2-RTS	LCD1-DATA22	PJ22	23	58	PG5	SDC1-D3	
UART2-CTS	LCD1-DATA23	PJ23	24	57	PG6	UART1-TX	
UART4-TX	LCD1-CLK	PJ24	25	56	PG7	UART1-RX	
UART4-RX	LCD1-DE	PJ25	26	55	PG8	UART1-RTS	
UART4-RTS	LCD1-HSYNC	PJ26	27	54	PG9	UART1-CTS	
UART4-CTS	LCD1-VSYNC	PJ27	28	53	PG10	I2S1-MCLK	
	LRADC0		29	52	PG11	I2S1-BCLK	
	LRADC1		30	51	PG12	I2S1-LRCK	
	GPADC3		31	50	PG13	I2S1-DIN1	
	GPADC4		32	49	PG14	I2S1-DOUT1	
	GPADC5		33	48	PM0	S-PWM2	
	GPADC6		34	47	PM1	S-PWM3	
	GPADC7		35	46	PM2	S-TWI1-SCL	
	GPADC8		36	45	PM3	S-TWI1-SDA	
	GPADC9		37	44	PM4	S-TWI2-SCL	
	GPADC10		38	43	PM5	S-TWI2-SDA	
	GPADC11		39	42			
	GPADC12		40	41			GND

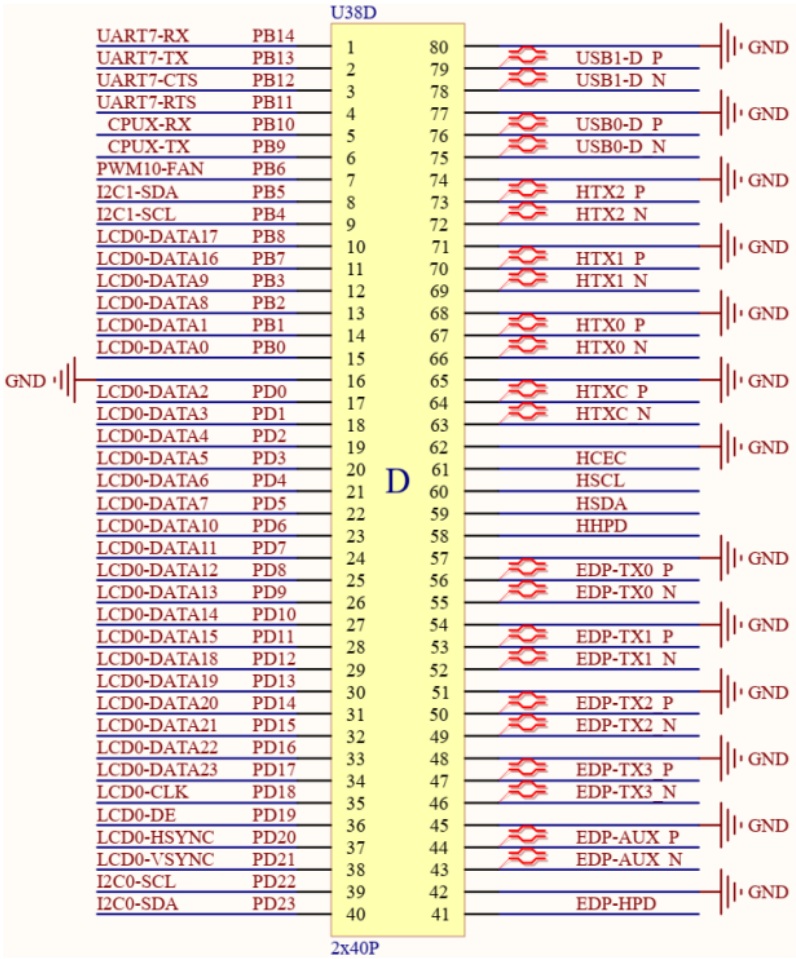
Core Board Connector A PINOUT



Core Board Connector B PINOUT



Core Board Connector C PINOUT



Core Board Connector D PINOUT

Only the functions multiplexed on our company's matching baseboard by the core board (i.e., the multiplexed functions in the factory system) are marked here.

To view the available pin multiplexing functions, please refer to *ATK-CLT527B Core Board Pin Multiplexing Table_V1.0.xlsx* or *T527_PINOUT_V0.91.xlsx*.

2.5 Number of Reusable Peripheral Interfaces

The core board leads out most of the IO pins of the T527 main control chip. Users can design their own baseboards according to actual needs to utilize the IO resources on the core board and multiplex the IO into required functions.

According to different peripheral interfaces, the maximum quantity of each reusable peripheral for the ATK-CLT527B core board is listed here. The specific selection can be combined with the chip datasheet.

(Maximum quantity of a single peripheral: refers to the maximum number of a certain peripheral that can be used by the core board without considering other peripherals.)

Peripheral Interface	Quantity	Description
Ethernet	≤2	- Supports 10/100/1000M data transfer rate - Supports RMII/RGMII interface
UART	≤10	- Maximum rate: 10 Mbit/s

		- Supports 5~8 data bits and 1/1.5/2 stop bits - Supports software/hardware flow control
SPI	≤4	- Supports configurable Master/Slave mode - Supports a maximum clock frequency of 100 MHz - In addition to standard SPI mode, it also supports DBI (Display Bus Interface) mode (SPI1 only) - Supports Single/Dual/Quad data modes (Dual/Quad not supported in Slave mode)
JTAG	≤4	- Supports ARM CPU/CPUS/DSP/RISC-V JTAG debugging
CAN	≤2	- Standard CAN interface, does not support CAN-FD
PWM	≤30-ch	- Supports pulse / period / complementary pair output - Supports input capture - Equipped with a programmable dead-time generator - Output frequency range: 0–24M / 100M. Duty cycle adjustable from 0% to 100%, with a minimum resolution of 1/65536
I2S/PCM	≤4	- Supports Master/Slave mode - Supports up to 14 channels for playback and recording - Hardware supports 8/12/16/20/24/28/32-bit data precision - Sampling rate supports 8 kHz ~ 384 kHz
TWI	≤8	- Compatible with standard I2C bus protocol - Supports 100 Kbit/s (Standard mode) and 400 Kbit/s (Fast mode) rates - Supports 7/10-bit device addressing modes
RGB	≤2	- Supports up to 1920×1080@60fps - Supports RGB888/RGB666/RGB565
HDMI TX	1	- Supports HDMI 2.0 protocol, up to 4K@60fps - Supports I2S
LVDS	≤4	- Supports up to 1366×768@60fps (Single-Link) and 1920×1080@60fps (Dual-Link) - Supports both VESA and JEIDA data formats
MIPI DSI	≤2	- Supports 2 channels of 4-lane MIPI DSI interface, supports Dual-Link - Supports up to 2560×1600@60fps (Dual-Link) and 1920×1200@60fps (Single-Link) - Supports MIPI DSI v1.02, MIPI DCS v1.01 and D-PHY v1.2 protocols
eDP	1	- Supports eDP 1.3 protocol, 4-channel eDP TX interface, up to 2.5K@60fps and 4K@30fps - Supports I2S

LRADC	2-ch	- 6-bit resolution, maximum sampling rate 2 KHz, analog input voltage range 0~1.8 V
GPADC	21-ch	- 12-bit resolution, maximum sampling rate 1 MHz, 10-bit sampling accuracy, analog input voltage range 0~1.8 V
SDIO	≤1	- Compliant with SDIO 3.0 protocol
SD Card	≤1	- Compliant with SD 3.0 protocol
SPIF	≤1	- Supports standard SPI - Part of the IO pins are multiplexed with eMMC, so full functions are not supported. Customers are advised to use with caution!
MIPI CSI	≤4	- Supports up to 4 channels of 2-lane MIPI CSI or 2 channels of 4-lane MIPI CSI - Supports up to 8M@30fps input, with resolution up to 3264(H)×2448(V)
Parallel CSI	≤1	- 16-bit parallel CSI interface - Supports 8/10/12/16-bit width - Supports BT.656, BT.601, BT.1120 interface standards - Maximum pixel clock: 148.5 MHz
PCIe	≤1	- PCIe 2.1 x1 lane, 5 Gbps per lane - IO multiplexed with USB 3.1 - Supports RC mode only
IR RX	≤2	- Supports NEC format
IR TX	≤1	- Supports firmware programming and ADB debugging - Supports High-Speed (480 Mbps), Full-Speed (12 Mbps) and Low-Speed (1.5 Mbps)
USB2.0 OTG	1	- Supports firmware programming and ADB debugging - Supports High-Speed (480 Mbps), Full-Speed (12 Mbps) and Low-Speed (1.5 Mbps)
USB2.0 HOST	1	- Supports High-Speed (480 Mbps), Full-Speed (12 Mbps) and Low-Speed (1.5 Mbps).
USB3.1 OTG	≤1	- USB 3.1 and PCIe share IO pins; there is no conflict if only USB 2.0 is used. - Supports Super-Speed (5 Gbps), High-Speed (480 Mbps), Full-Speed (12 Mbps) and Low-Speed (1.5 Mbps).
Audio Codec	1	- The T527 chip integrates an Audio Codec. - Supports 3 channels of audio output: LINEOUTLP/N, LINEOUTRP/N, HPOUTL/R. - Supports 3 channels of audio input: MICIN1P/N, MICIN2P/N, MICIN3P/N.
DMIC	≤1	- Supports up to 4 digital PDM microphones - Supports sampling rates from 8 KHz to 48 KHz

OWA RX	≤1	- One Wire Audio, compatible with S/PDIF interface
OWA TX	≤1	- One Wire Audio, compatible with S/PDIF interface
GPIO	≤189	- Note: Function multiplexing.

2.6 Software Parameters

The core board supports running different operating systems and various drivers, as shown in the table below.

System and Driver Support		
Operating System	Linux Buildroot (Default factory system, recommended)	
	Android13	
	Debian (Version included in Allwinner SDK, not maintained)	
GUI	QT 5.15.8	
	LVGL 8.3.2	
Driver Support	PMIC	eMMC
	LPDDR4X	ISP
	ADC	GPU
	LED	RTC
	KEY	I2C
	CAN	SPI
	I2S	UART
	RS485	USB2.0 HOST
	USB2.0 OTG	HDMI OUT
	USB3.0 OTG	eDP
	AUDIO OUT	MIC IN
	PWM	SD Card
	4G	5G RedCap
	WiFi	Bluetooth
	MIPI Camera	Touch Screen
	MIPI Display	LVDS display screen
	FAN	Ethernet
Note	Provides Tina5.0 Linux and Android 13 SDK source code packages for easy use and supports secondary development.	

Chapter 3. Electrical Characteristics

The electrical characteristics parameters of the core board are shown in the table below.

Operating Parameters	Minimum Value	Typical Value	Maximum Value
Core Board Operating Voltage	/	4V	/
Core Board Operating Temperature (Industrial Grade)	-40℃	/	+85℃

The reference power consumption values of the core board are shown in the table below.

No.	Test Case	Input Power of Test Board	Typical Current	Typical Power Consumption
1	Core board powered on, no system running.	12V	80mA	0.96W
2	The core board only runs the factory system without connecting any external modules.	12V	205mA	2.46W

Notes:

1. The above reference power consumption values are measured on the ATK-DLT527B development board with a DC input power supply of 12V/2.5A.

2. The core board is equipped with multiple interfaces. Actual power consumption is related to the number of peripherals used, external modules, and other factors, and should be measured according to actual application scenarios. The data provided here is for reference only.

Chapter 4. Mechanical Structure

4.1 PCB Parameters

Parameters	Core board
PCB Size	60mm*50mm
PCB Thickness	1.6mm
PCB Layers	10 layers
PCB Technology	Gold plating
Number of Mounting Holes	4 pieces

4.2 Core Board Mechanical Dimensions

The dimensions of the ALIENTEK ATK-CLT527B core board are 60mm*50mm. The mechanical drawing is shown in the figure below.

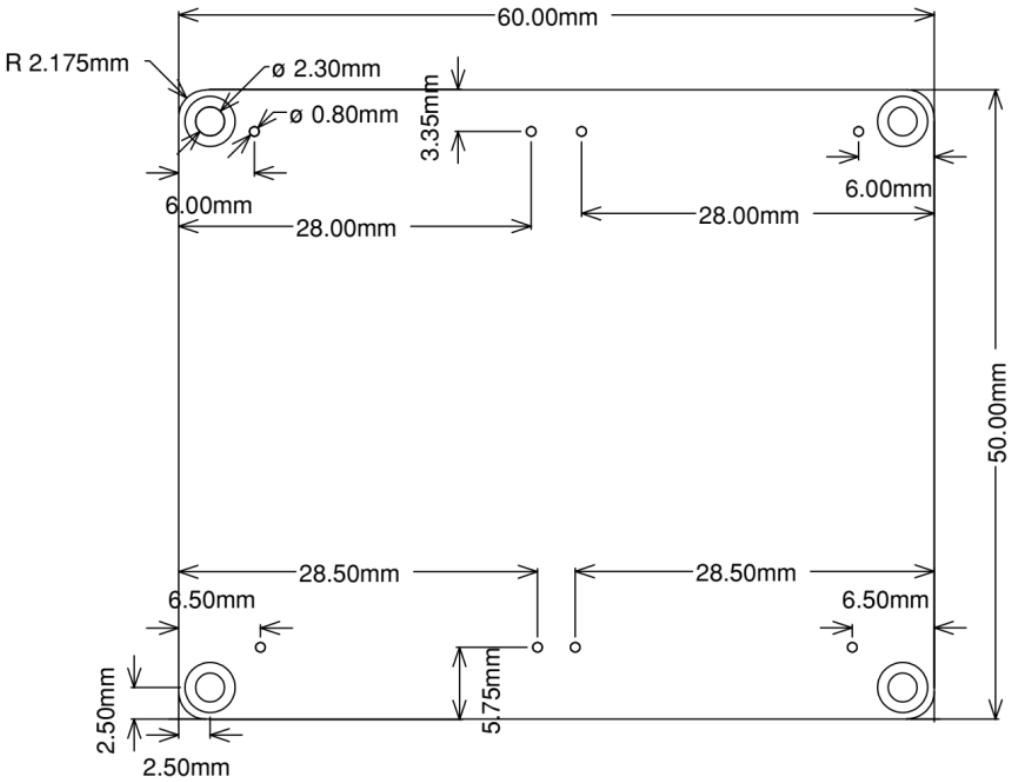


Figure 4.2-1 Core Board Mechanical Drawing

Note: The original file path of the core board mechanical drawing in the above figure is:

Development Board NetDisk Drive A - Basic Documents → 03. Core Board Documents → 03. Core Board Mechanical Structure → ATK-CLT527B V1.0 Mechanical Drawing.pdf

The core board and base board are connected via a BTB (board-to-board) connector. The mating height of the connector is 3mm. After all components are soldered and assembled on the core board PCB, the maximum overall height of the PCBA is approximately 6.3mm (measured at the tallest component).

Chapter 5. Core Board Resources

ALIENTEK provides abundant software resources for the ATK-CLT527B core board, including but not limited to Tina5.0 Linux SDK, Android 13 SDK, Linux Buildroot & Android 13 system firmware, Qt source code, relevant development tools and development environments, etc. Corresponding development documents are also provided, covering development environment setup, SDK compilation, packaging & burning instructions, hardware testing guides, hardware design references, etc.

In addition, ALIENTEK has specially compiled a series of materials for core board users that will be used during the development phase, including base board schematics, base board design references, mechanical structures, component footprints, connector specifications, Linux Buildroot & Android system firmware and source code, compilers, software packages, etc., to facilitate rapid development and improve development efficiency.

All supporting resources of the development board are available for download via cloud disk.

5.1 Resource Download

ATK-CLT527B Core Board Resource Download Center:

http://www.openedv.com/docs/boards/arm-linux/T527hxb%20Linux.html

ATK-DLT527B Development Board Resource Download Center (It is recommended to download the development board resources, which already include the core board resources):

http://www.openedv.com/docs/boards/arm-linux/T527%20Linux.html

5.2 Resource Description

The level-1 directory of the development board resources on NetDisk Drive A in Chinese version is shown in the figure below.

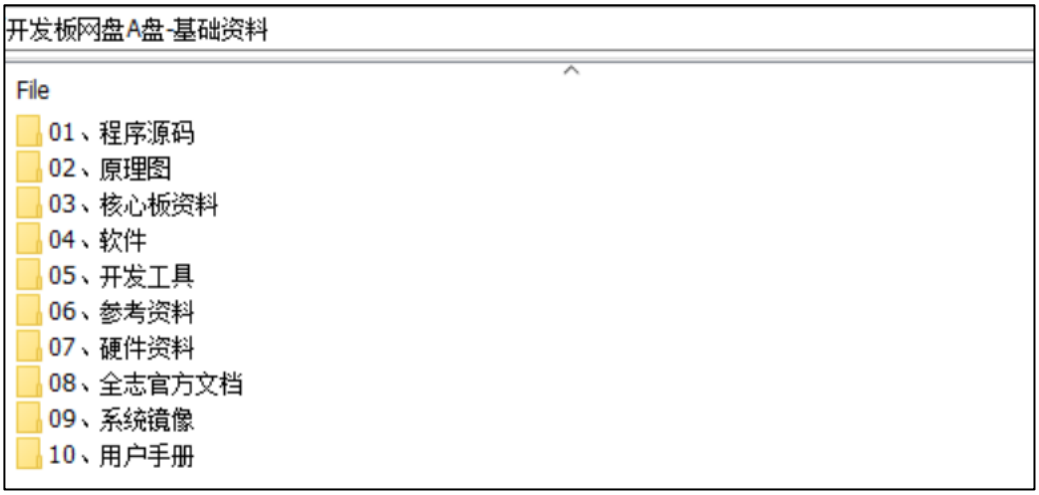


Figure 5.2-1 Level-1 Directory of Development Board Drive A Resources

For English resources, please refer to the corresponding development board repository on GitHub:

https://github.com/openedv/development_board_ATK-DLT527B.git

Directory	Description
01_codes	Source code collection, e.g., complete Qt routine source code

02_sch	Development Board Base Board Schematic
03_core_board_reference	Includes core board-related materials: core board pin definitions, onboard chip documentation, core board mechanical structure, baseboard design references, specifications, certification documents, package libraries, etc.
04_softwares	Collection of required software tools, including serial terminal software, FTP client software, USB-to-serial driver, VMware virtual machine software, etc.
05_tools	Collection of development tools, including cross-compiler, ADB debugging tool, programming tool, Allwinner official development tools, etc.
06_reference_data	Reference materials collection, including protocol manuals, ARM manuals, and other reference documents.
07_hardware	Hardware-related materials collection, including baseboard onboard chip documents, development board specifications, development board package libraries, development board mechanical dimensions, certification documents, etc.
08_allwinner_reference	Includes development documents provided by Allwinner official, such as hardware documents, software files, tool files, etc.
09_system_image	Includes Linux Buildroot system firmware and Android 13 system firmware. The development board comes pre-installed with Linux Buildroot at the factory.
10_user_manual	Stores the documents provided to users by ALIENTEK. However, this directory is empty as the contents have been moved to the Yuque online documentation.

Note: Due to limited space, the corresponding directory contents are not elaborated here. Please download the development board resources for detailed inspection.

The level-1 directory of the development board NetDisk Drive B in Chinese version is shown below:

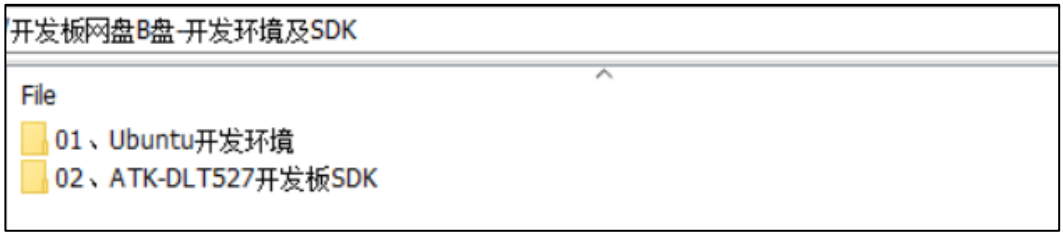


Figure 5.2-2 Level-1 Directory of Development Board Drive B Resources

Development Board Drive B Resource Directory Description:

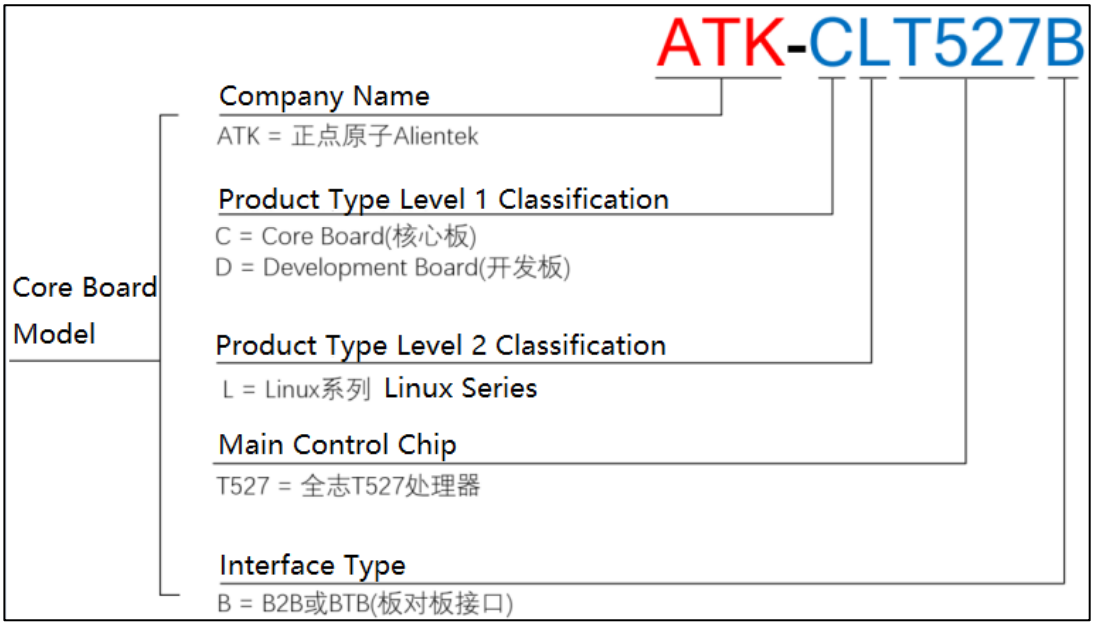
Directory	Description
01. Ubuntu Development Environment	To facilitate customers' rapid development, our company provides an Ubuntu virtual machine image with a pre-built basic development environment. Customers can directly use this Ubuntu virtual machine for software

	development on the T527 platform, without the need to build an Ubuntu development environment from scratch.
02. ATK-DLT527 Development Board SDK	Supporting SDK, including Tina5.0 Linux SDK and Android13 SDK. Users can carry out secondary development based on the SDK.

Chapter 6. Core Board Selection

6.1 Model Naming Convention

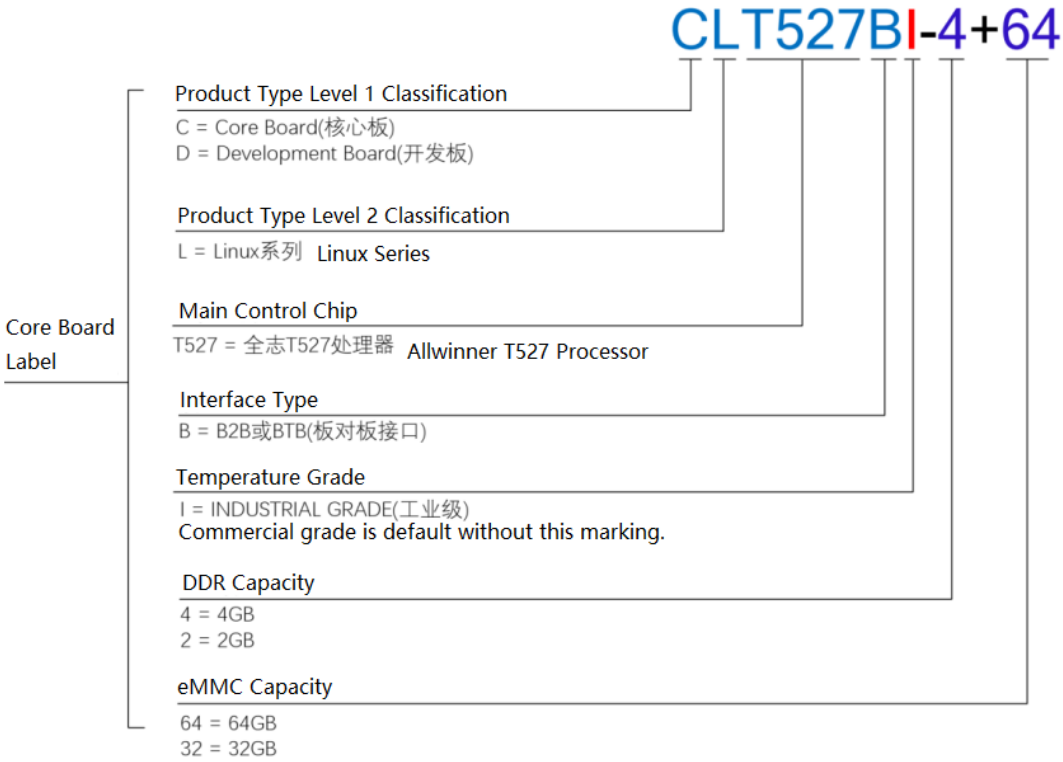
The core board model is ATK-CLT527B, and the silk-screen label on the core board is CLT527B. The model naming convention is shown below:



6.2 Label Naming Convention

To distinguish the storage capacity and operating temperature range of the core board, each core board is affixed with a label marked with a name before factory shipment.

The label naming convention is shown below:



6.3 Purchase Configurations

According to the storage capacity and operating temperature range, we provide the following 4 purchase configurations:

	Configuration 1	Configuration 2	Configuration 3	Configuration 4
DDR	2GB	2GB	4GB	4GB
eMMC	32GB	32GB	64GB	64GB
Temperature Range Grade	Commercial Grade (0°C ~ +70°C)	Industrial Grade (-40°C ~ +85°C)		
Sticker Naming	CLT527B-2+32	CLT527BI-2+32	CLT527B-4+64	CLT527BI-4+64

Please note that commercial-grade core boards are **not currently available for sale** on our platform.

Customers can **only purchase industrial-grade core boards** at this time.

For details, please consult our business team!

6.4 T527 Chip Model

There are two specific models in the T527 chip series:

T527M02X0DCH and **T527M00X0DCH**.

The difference between them lies in whether an NPU unit is integrated:

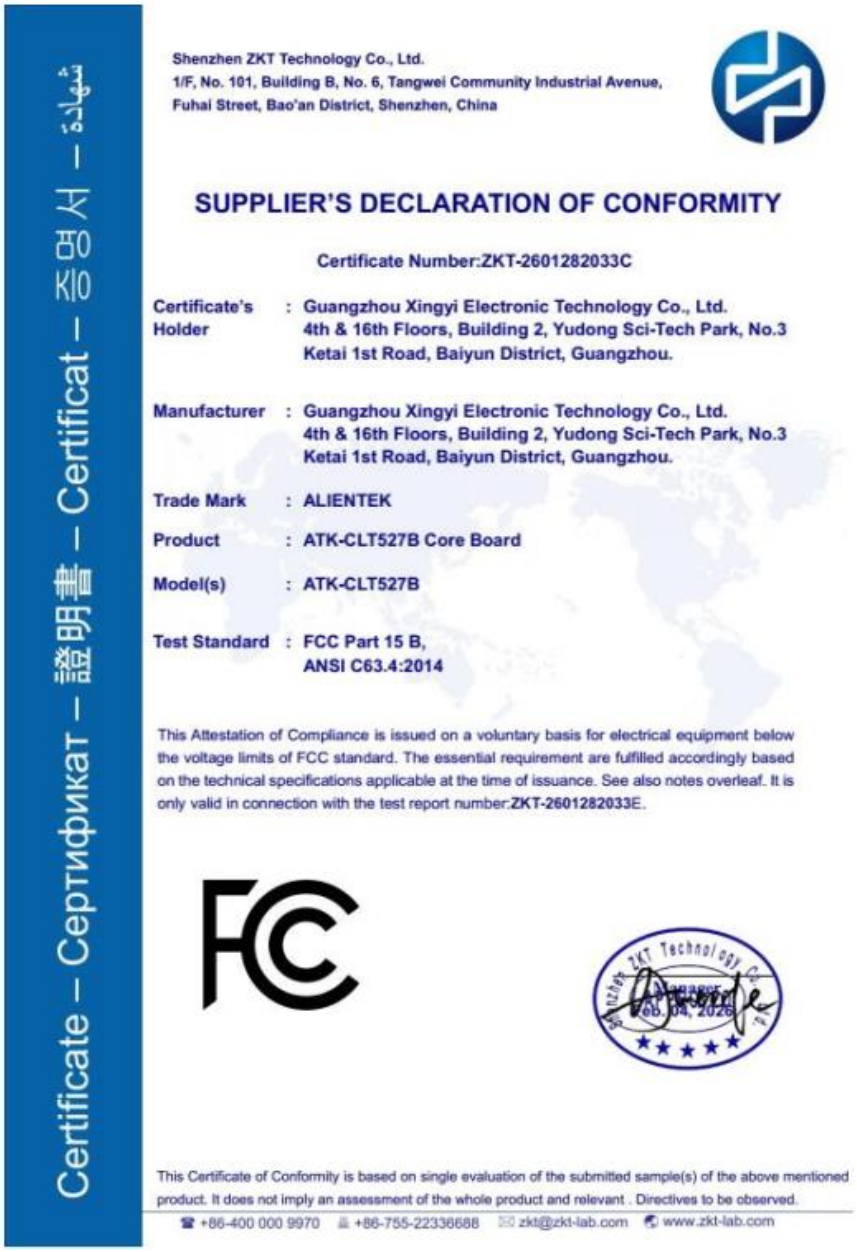
- **T527M02X0DCH**: Supports a 2 TOPS NPU computing unit.
- **T527M00X0DCH**: Does not have a built-in NPU computing unit.

Orderable Device	NPU	Package
T527M02X0DCH	Support	17mm * 17mm, FCBGA 664 balls
T527M00X0DCH	Not Support	17mm * 17mm, FCBGA 664 balls

It should be noted that all core boards currently sold by our company are equipped with the T527 main control chip model **T527M00X0DCH**, which is the version without NPU.

Chapter 7. Core Board Certification Instructions

7.1 FCC Certification



Certificate – Сертификат – 證明書 – Certificat – 증명서 – شهادة

Shenzhen ZKT Technology Co., Ltd.
1/F, No. 101, Building B, No. 6, Tangwei Community Industrial Avenue,
Fuhai Street, Bao'an District, Shenzhen, China

SUPPLIER'S DECLARATION OF CONFORMITY

Certificate Number:ZKT-2601282033C

Certificate's Holder : Guangzhou Xingyi Electronic Technology Co., Ltd.
4th & 16th Floors, Building 2, Yudong Sci-Tech Park, No.3
Ketai 1st Road, Baiyun District, Guangzhou.

Manufacturer : Guangzhou Xingyi Electronic Technology Co., Ltd.
4th & 16th Floors, Building 2, Yudong Sci-Tech Park, No.3
Ketai 1st Road, Baiyun District, Guangzhou.

Trade Mark : ALIENTEK

Product : ATK-CLT527B Core Board

Model(s) : ATK-CLT527B

Test Standard : FCC Part 15 B,
ANSI C63.4:2014

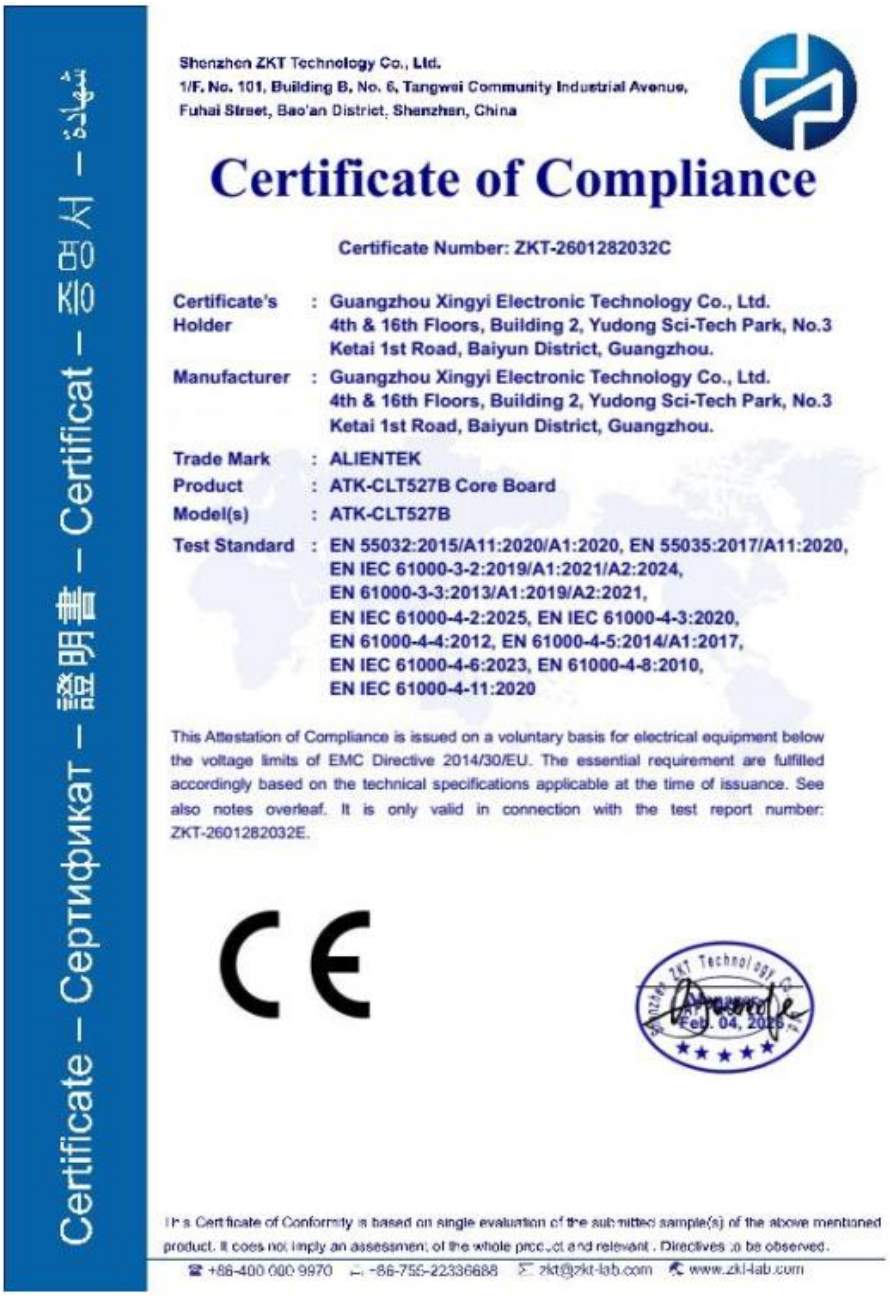
This Attestation of Compliance is issued on a voluntary basis for electrical equipment below the voltage limits of FCC standard. The essential requirement are fulfilled accordingly based on the technical specifications applicable at the time of issuance. See also notes overleaf. It is only valid in connection with the test report number:ZKT-2601282033E.

This Certificate of Conformity is based on single evaluation of the submitted sample(s) of the above mentioned product. It does not imply an assessment of the whole product and relevant . Directives to be observed.

+86-400 000 9970 +86-755-22336688 zkt@zkt-lab.com www.zkt-lab.com

Figure 7.1-1 FCC Certification Certificate

7.2 CE Certification



Certificate – Сертификат – 證明書 – Certificat – 증명서 – شهادة

Shenzhen ZKT Technology Co., Ltd.
1/F, No. 101, Building B, No. 6, Tangwei Community Industrial Avenue,
Fuhai Street, Bao'an District, Shenzhen, China

Certificate of Compliance

Certificate Number: ZKT-2601282032C

Certificate's Holder	:	Guangzhou Xingyi Electronic Technology Co., Ltd. 4th & 16th Floors, Building 2, Yudong Sci-Tech Park, No.3 Ketai 1st Road, Baiyun District, Guangzhou.
Manufacturer	:	Guangzhou Xingyi Electronic Technology Co., Ltd. 4th & 16th Floors, Building 2, Yudong Sci-Tech Park, No.3 Ketai 1st Road, Baiyun District, Guangzhou.
Trade Mark	:	ALIENTEK
Product	:	ATK-CLT527B Core Board
Model(s)	:	ATK-CLT527B
Test Standard	:	EN 55032:2015/A11:2020/A1:2020, EN 55035:2017/A11:2020, EN IEC 61000-3-2:2019/A1:2021/A2:2024, EN 61000-3-3:2013/A1:2019/A2:2021, EN IEC 61000-4-2:2025, EN IEC 61000-4-3:2020, EN 61000-4-4:2012, EN 61000-4-5:2014/A1:2017, EN IEC 61000-4-6:2023, EN 61000-4-8:2010, EN IEC 61000-4-11:2020

This Attestation of Compliance is issued on a voluntary basis for electrical equipment below the voltage limits of EMC Directive 2014/30/EU. The essential requirement are fulfilled accordingly based on the technical specifications applicable at the time of issuance. See also notes overleaf. It is only valid in connection with the test report number: ZKT-2601282032E.

CE

Approved
Feb. 04, 2026

This Certificate of Conformity is based on single evaluation of the submitted sample(s) of the above mentioned product. It does not imply an assessment of the whole product and relevant Directives to be observed.

☎ +86-400 000 9970 +86-755-22336688 zkt@zkt-lab.com www.zkt-lab.com

Figure 7.2-1 CE Certification Certificate

7.3 RoHS Certification

Certificate – Сертификат – 證明書 – Certificat – 증명서 – شهادة

Shenzhen ZKT Technology Co., Ltd.
1/F, No. 101, Building B, No. 6, Tangwei Community Industrial Avenue,
Fuhai Street, Bao'an District, Shenzhen, China

Certificate of Compliance

Certificate Number: ZKT-2601282034C

Certificate's Holder : Guangzhou Xingyi Electronic Technology Co., Ltd.
4th & 16th Floors, Building 2, Yudong Sci-Tech Park, No.3 Ketai 1st Road, Baiyun District, Guangzhou.

Manufacturer : Guangzhou Xingyi Electronic Technology Co., Ltd.
4th & 16th Floors, Building 2, Yudong Sci-Tech Park, No.3 Ketai 1st Road, Baiyun District, Guangzhou.

Product : ATK-CLT527B Core Board

Trade Mark : ALIENTEK

Model(s) : ATK-CLT527B

Test Method : IEC 62321-3-1:2013, IEC 62321-4:2013+A1:2017, IEC 62321-5:2013, IEC 62321-6:2015, IEC 62321-7-2:2017, IEC 62321-7-1:2015, IEC 62321-8:2017.

Test Report No. : ZKT-2601282034R

The EUT described above has been consolidated by us and found in compliance with the council RoHS Directive 2011/65/EU Annex II amending Annex (EU)2015/863. It is possible to use CE marking to demonstrate the compliance with this RoHS Directive. It is only valid in connection with the test report

After preparation of the necessary technical documentation according the Requirement of EN IEC 63000:2018 as well as the verification of conformity the CE marking as shown below can Be affixed on the equipment.

RoHS

Manager
Feb. 03, 2026

This Certificate of Conformity is based on single evaluation of the submitted sample(s) of the above mentioned product. It does not imply an assessment of the whole product and relevant Directives to be observed.

+86-400 000 9970 +86-755-22336688 zkt@zkt-lab.com www.zkt-lab.com

Figure 7.3-1 RoHS Certification Certificate

Chapter 8. Precautions and maintenance

Notes

- Do not plug and unplug peripheral modules with power!
- Before using the product, please carefully read this manual and related development manuals, and pay attention to the applicable matters of the platform.
- Follow all instructions and warnings on the product.
- Please use this product in a cool, dry and clean place.
- Please keep the product dry. If any liquid splashes or soaks, power off immediately and let dry thoroughly.
- Do not use organic solvents or corrosive liquids to clean the product.
- Do not use or store this product in dusty, dirty and messy environment.
- If not used for a long time, please package this product, pay attention to moisture-proof and dust-proof.
- Pay attention to the ventilation and heat dissipation of the product during use to avoid component damage caused by excessive temperature during operation.
- Do not use this product in alternating hot and cold environment to avoid dew damage to components.
- Do not treat this product roughly, drop, knock or shake violently may damage the line and components.
- Pay attention to anti-static when using this product.
- FPC flexible cable is fragile, when plugging cable, pay attention to check whether the metal at both ends of the cable is misplaced and falling off.
- All products have passed the product test before shipment. Please use the development board corresponding to the ALIENTEK for power on test for the first time.
- Do not repair or disassemble the company's products by yourself. If the product fails, please contact the company in time for maintenance.
- Unauthorized modification or use of unauthorized parts may damage the product, the resulting damage will not be repaired.

Chapter 9. After sales service

9.1 Terms of after-sales service

1). After receiving the goods, please open them in front of the express, and sign after acceptance. If you find that the goods are less after signing, take photos in time and contact the seller's customer service to explain the situation within 15 days. If the feedback is lack of goods after 15 days, we will not reissue the goods. Other reasons notwithstanding).

2). 15 days -1 month: we are responsible for the return freight repair of product problems. Human factors damage expensive main chip or LCD screen, touch screen. The buyer needs to pay the cost and one time shipping fee, no maintenance fee.

3). 1-3 months: the problem of the product itself (non-human factors), we are responsible for the delivery of the past freight maintenance. If the main chip is burned out and the LCD screen and touch screen are damaged, the buyer needs to pay the cost, and the maintenance fee is not charged.

4) After 3 months: the buyer shall bear the return freight and the cost of chip, LCD screen and touch screen. No service charge.

9.2 After-sales Support

Technical support:

QQ group: ALIENTEK Allwinner Communication group: 1032186104

ALIENTEK Allwinner User Group: 2152038686 (order number required)

Taobao shop: ALIENTEK flagship store

Business Contact: **ALIENTEK Flagship Store**, transfer to Business Customer Service

Tel: 020-38271790 (reception transfers to Linux Technical Support)

Forum: http://www.openedv.com/forum.php